Case 47: Urban-to-wildland WU-E transition and isolated wildland receivers

1 Verification purpose

This case tests the urban-to-wildland interface of the Wildland–Urban ELMFIRE (WU-E) model. It separates three observables that can otherwise be mistaken for one another: heat emitted by an urban source, heat accumulated at a wildland receiver, and arrival of the level-set fire front at that receiver. The reviewed WU-E components are identical to source revision `a2dfbcdf72209733c000e5d3431e44723e15efea`; this identity claim does not extend to other modified model components. The processes examined are the building design-fire curve, urban heat transfer, fuel-dependent velocity selection in the level-set formulation, and ordinary Rothermel surface-fire spread.

Two urban sources are initialized in otherwise nonburnable surroundings. One illuminates a disconnected wildland receiver; the other starts a three-cell wildland path. The factorial experiment varies urban peak heat-release rate (HRR) per unit area (HRRPUA) over 100 kW m^{-2} and 400 kW m^{-2}, wildland spread-rate adjustment factor over 0 and 1, and replaces the middle path cell by a nonburnable barrier. Terrain is flat and deterministic; firebrands, crown fire, wind fluctuations, and suppression are disabled. This is numerical verification and implementation characterization, not physical validation.

The precise verification question is: across all eight simulation conditions, does the pinned implementation preserve the fourfold source-HRR and received-heat responses, keep the disconnected heat-exposed wildland cell unignited, expose ordinary wildland velocity to spread-rate adjustment factor, and preserve the nonburnable barrier? The separate publication claim that heat alone ignites a wildland target is reported as **NOT EVALUABLE** because the pinned source contains no corresponding target-state transition.

Current result: NOT EVALUABLE. 8 of 8 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

Urban-to-wildland coupling is outside the downscaled building-to-building baseline used by CASE43–46 and CASE48–49. It is retained as a separate published capability [1, 2]. The proceedings article (pages 2–3, equations (2)–(8)) and the supplied thesis (PDF pages 63–67, printed pages 41–45, equations (2.81)–(2.95)) describe source heat transfer using

$$\dot{q}_c = \frac{\mathrm{HRR}\, A_t}{\Delta x^2}, \qquad \dot{q}''_r = \frac{0.3\,\mathrm{HRR}}{4\pi R^2}, \tag{1}$$

$$\dot{q}_t = \alpha_c \dot{q}_c + \alpha_r \dot{q}''_r \Delta x^2, \qquad E_t^{n+1} = E_t^n + \dot{q}_t^n \Delta t. \tag{2}$$

Here A_t is the flame-overlap area, R is source–target distance, α_c is a combustible-area coefficient, and α_r includes target absorptivity. The documented ignition idealization is $E_t \geq \mathrm{FTP}\,\Delta x^2$, giving $U = \sum \dot{q}_t/(\mathrm{FTP}\,\Delta x)$.

The implementation performs a related discrete heat calculation. For every burning source it scans the configured interaction-band thickness, computes ellipse-overlap factors, sums transient direct flame contact (DFC) and radiation at each target, and accumulates both after multiplication by $\Delta t \Delta x^2$. Target coefficients come from the building-fuel table even where the target fire behavior fuel model (FBFM) is wildland. These arrays are observable as transient heat flux and terminal total-received-heat rasters.

That heat state is not, however, consumed by a wildland ignition law at the pinned revision. The WU-E heat-to-velocity routine is selected only when the current target has fuel model 91. A target with

a fuel model other than 91 instead follows the ordinary surface-fire path; for Rothermel its no-wind base velocity contains the explicit multiplier

$$V_{s,0} = \text{ADJ } D \; I_R \xi / (\rho_b \epsilon Q_{ig}), \tag{3}$$

where D is the diurnal adjustment. Consequently, received WU-E heat and Rothermel front arrival are distinct state variables in this experiment.

3 Derivation of expected behavior

Let the low design-fire curve be $H_L(t) = H_0 h(t)$, where the common piecewise-linear shape h grows through 60 s, remains fully developed through 300 s, and decays to zero at 600 s. The high table changes only the amplitude, so

$$H_H(t) = 4H_L(t). \tag{4}$$

The heat-loop geometry and target coefficients are unchanged between a matched low/high pair. Both direct flame contact and radiation are linear in instantaneous HRR; therefore, at the disconnected receiver,

$$\frac{\max H_H}{\max H_L} = 4, \qquad \frac{E_{t,H}}{E_{t,L}} = 4. \tag{5}$$

Raster roundoff motivates small relative tolerances rather than exact bitwise equality.

The limits are preregistered numerical discriminators, not experimental uncertainties. The 1-percent source-ratio limit is generous relative to float32 storage while still detecting a material departure from fourfold scaling; the 0.1-percent repeat limit is tighter because the paired source inputs are identical. The 2-percent accumulated-heat ratio allows repeated float32 additions over 600 one-second updates. A 1-s time of arrival (TOA) difference is exactly one configured solver step. Finally, 10^{-4} m/min would move a front only 0.001 m over the full run, far below a 10-m cell, so it separates a dynamically positive path from numerical zero without serving as a physical rate of spread (ROS) threshold.

The heat experiment places its wildland receiver three cells downwind and separates it from the source with two FBFM99 cells. It lies inside the five-cell WU-E scan but cannot be reached by the continuous front. The pinned algorithm accumulates positive heat there but has no conditional assignment that changes its arrival time from that heat; its time of arrival must therefore remain nodata. This is an implementation invariant, not acceptance of the published capability as correct.

In the separate path experiment, FBFM1 cells carry spread-rate adjustment factor=0 or 1. For spread-rate adjustment factor=0 their ordinary Rothermel base velocity is zero; for spread-rate adjustment factor=1, the configured dry fuel and 20 mph westerly wind give a positive value. The open path therefore supplies a direct control on ordinary surface spread. Replacing its middle cell by FBFM99 supplies an independent nonburnable invariant. No arrival at that named FBFM99 barrier cell is permitted; the metric does not generalize this single-cell control to every background pixel.

4 Verification metrics and acceptance criteria

The evaluator first verifies prepared-input hashes, all eight identities, a unique dump-time simulation specification per simulation condition, a unique final record exactly at 600 s, and grid agreement for every required GeoTIFF. It requires terminal time of arrival, spread rate, fireline intensity, total direct flame contact, and total radiation, plus complete HRR/DFC/radiation transient series matching every dump-simulation specification index on the exact 60 s schedule. With feedback enabled, it also requires exactly 600 logged solver-step timestamps on the configured 0–599 s sequence before selecting the

explicit post-step final dump at 600 s. Any missing, duplicate, malformed, off-grid, or nonterminal evidence makes the run **NOT EVALUABLE**; process exit status is never evidence of success.

Expected property	Observable and metric	Sensitivity and selection rationale	Component limit
Fourfold source amplitude	Maximum of the two source cells over every HRR transient; matched high/low ratio error	Detects table selection, design-curve, or output errors without depending on front shape	relative error ≤ 0.01
Control independence	Range of source peaks across ADJ/barrier controls at fixed amplitude	Detects unintended coupling of receiver controls into the source curve	relative range ≤ 0.001
Heat reaches isolated cell	Terminal DFC plus radiation at the exact receiver cell	Proves absence of TOA is not explained by zero incident heat	minimum > 0
Linear heat response	Matched high/low ratio of isolated total heat	Detects nonlinear or incorrect HRR accumulation while holding geometry fixed	ratio error ≤ 0.02
No current heat-only target transition	Count of finite isolated-receiver TOA values	Directly tests the source-derived current invariant	0 of 8
Nonburnable control	Count of finite barrier-cell TOA values in closed simulation conditions	Detects leakage through the FBFM99 control	0 of 4
Ordinary surface control	Named-cell TOA plus finite terminal VS over FBFM1 cells	Distinguishes the ADJ-scaled Rothermel path from received heat without coercing nodata VS to zero	no adjacent/distal ADJ0 arrivals; all ADJ1-open arrivals; VS $> 10^{-4}$ m min^{-1}
Ignition-time control	Adjacent and distal TOA paired across the two open-path HRR simulation conditions	Detects accidental dependence of ordinary spread on the disconnected HRR-amplitude control	maximum difference ≤ 1 s

All current-implementation components must pass for the implementation characterization to be **PASS**. The intended heat-only urban-to-wildland capability remains **NOT EVALUABLE** independently, so the case-wide verification decision cannot be **PASS** at this source revision.

5 Simulation configuration and predicted outcome

The domain is a 310 m square in EPSG:32610, with north-up 10 m cells and nodata -9999. PHI is a clipped signed-distance field with exactly two negative urban-source cells and no cell centered on PHI zero. Both sources use building fuel model 1; every wildland receiver is FBFM1 and the background/barrier is FBFM99. Source separation exceeds twice the WU-E heat band, so the isolated receiver is influenced only by its designated heat source. A dormant scheduled ignition at 601 s prevents an unrelated stalled-front shortcut but occurs after the stop time.

Parameter	Configured value	Units	Role
Grid/domain	31×31 at 10; EPSG:32610	m/cell	Resolves source, receiver, and barrier cells exactly
Urban peak HRRPUA	100, 400	kW m^{-2}	Fourfold source and heat-amplitude pair
Wildland adjustment	0, 1	—	Turns ordinary Rothermel velocity off/on
Path control	FBFM1 or FBFM99 at middle cell	code	Open versus nonburnable barrier
Weather/terrain	20 mph from 270°; flat	mph, degree	Downwind heat exposure with fixed geometry
Building geometry	linear dimension 10, separation 10, footprint fraction 1	m, m, fraction	Fixed WU-E ellipse inputs
Time integration	$\Delta t = \Delta t_{\max} = 1$, $t_{stop} = 600$	s	Exactly 600 logged steps and an explicit post-step exact-stop final dump
Output cadence	60	s	Resolves rise, plateau, and decay
Disabled processes	spotting, crown fire, suppression, fluctuations	–	Removes alternate nonlocal or stochastic ignition

Figure 1 is generated only from the prepared case-specific GeoTIFFs. The predicted result is positive isolated-cell heat with no isolated TOA, a fourfold low/high response, zero ordinary wildland speed at ADJ0, positive ordinary speed for the open ADJ1 controls, and no barrier-cell TOA.

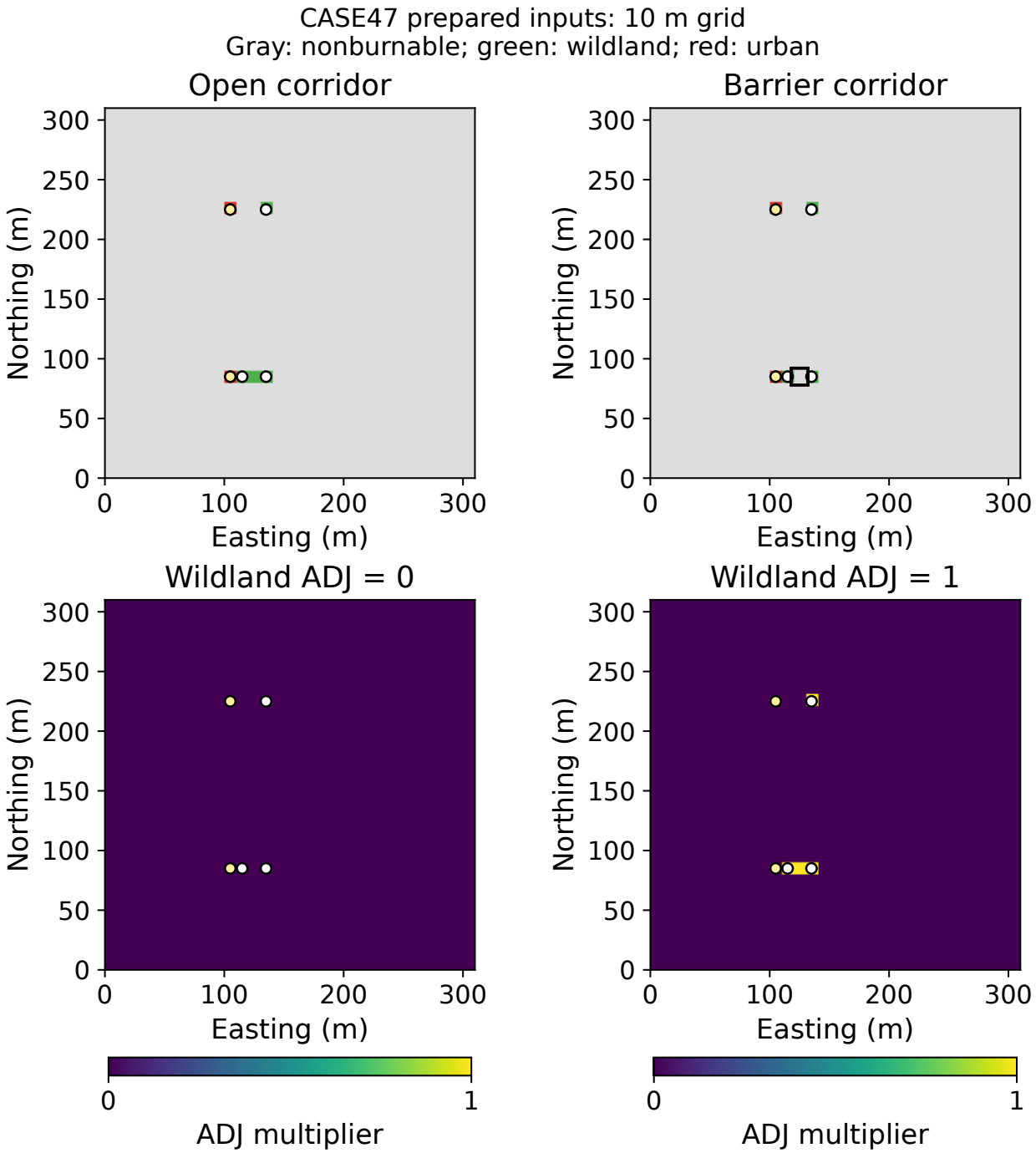


Figure 1: Whole-domain prepared inputs for representative high-HRR simulation conditions. FBFM91 marks each urban source, FBFM1 marks wildland receivers/path cells, FBFM99 is nonburnable, and the black contour is the generated initial PHI interface. Yellow circles mark urban sources; white circles mark wildland receivers. The upper row of markers is the isolated heat experiment; the lower row is the surface-spread experiment, with a square marking the barrier cell.

6 Actual simulation results

The authoritative record reports analysis procedure state **COMPLETE** and contains 8 of 8 required simulation conditions. The scientific decision is **NOT EVALUABLE**. Reason: Current behavior is characterized separately; the documented heat-only U-to-W capability has no executable reference calculation path at this source revision. The executed binary SHA-256 is `b7c9bb4dc72f01cd2c18e51 8b7dbd8fd76871728905e68423b103684f537a90a`; it is interpreted only against the pinned source reference calculation and is not assumed to equal it. Available results retain their actual output times and matching model, input, and simulation-record identities. All selected fields must belong to those completed simulations.

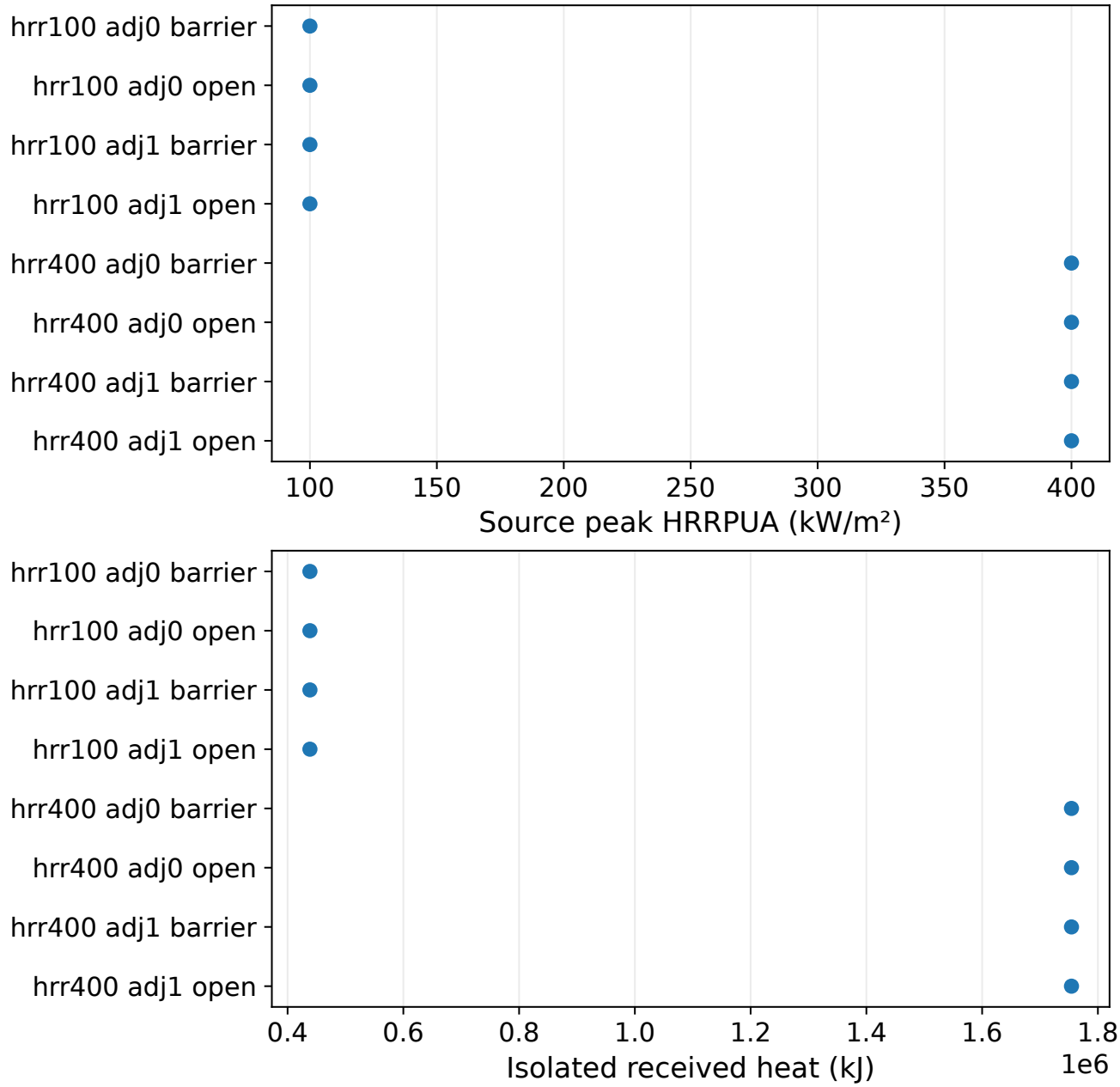


Figure 2: Actual case-specific source HRR and isolated-cell accumulated heat for all eight simulation conditions through 600 s. Values are read from ELMFIRE GeoTIFF outputs; they are not synthetic reference curves. Amplitude-ratio limits are 1% for source HRR and 2% for heat. Arrival times are shown separately.

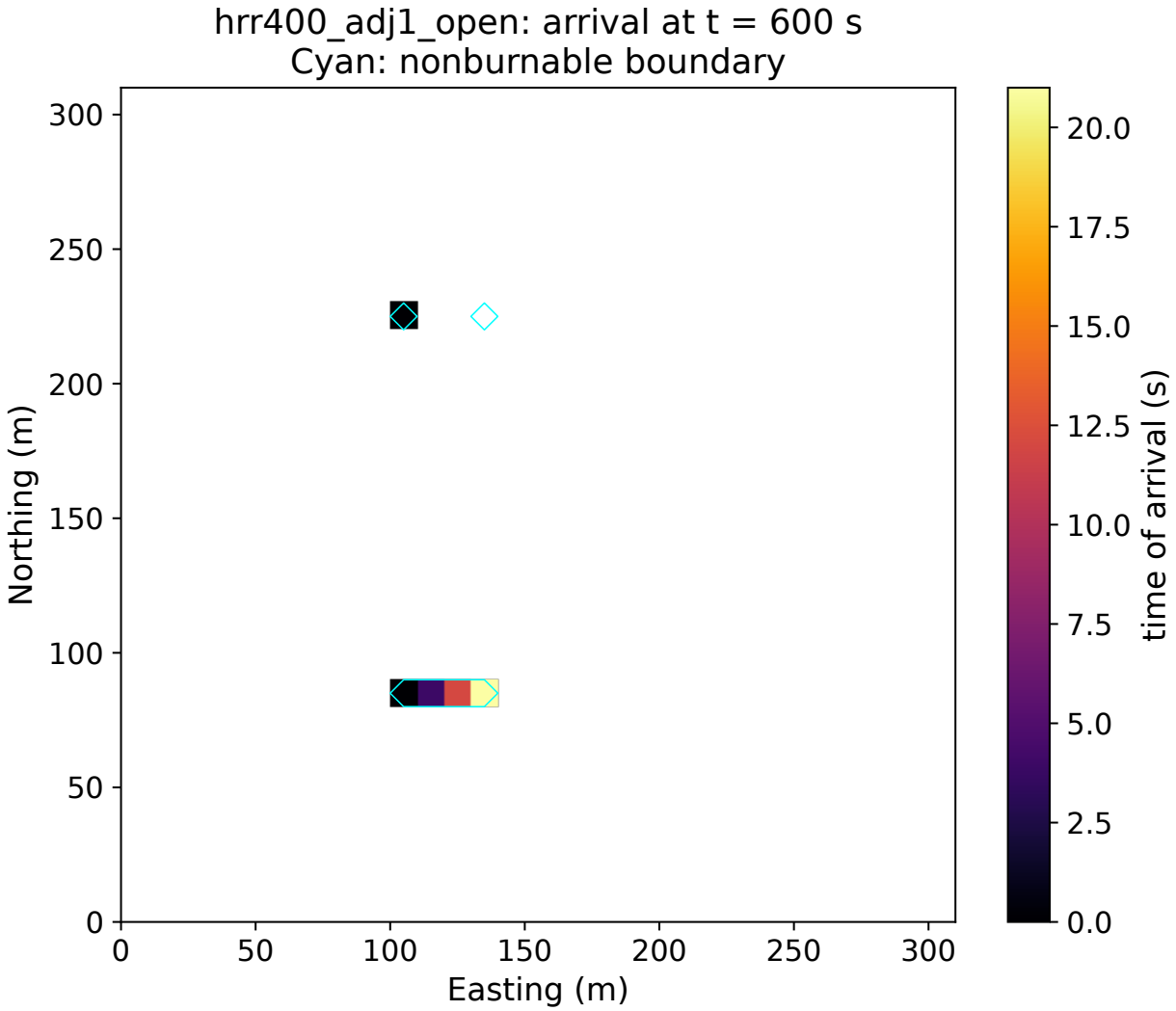


Figure 3: Actual whole-domain time of arrival for the open configuration with heat-release rate per unit area 400 and adjustment factor 1 at 600 s; nodata remains blank and the cyan line marks the FBFM99 boundary. The selected field is the unique terminal arrival-time result for this simulation condition.

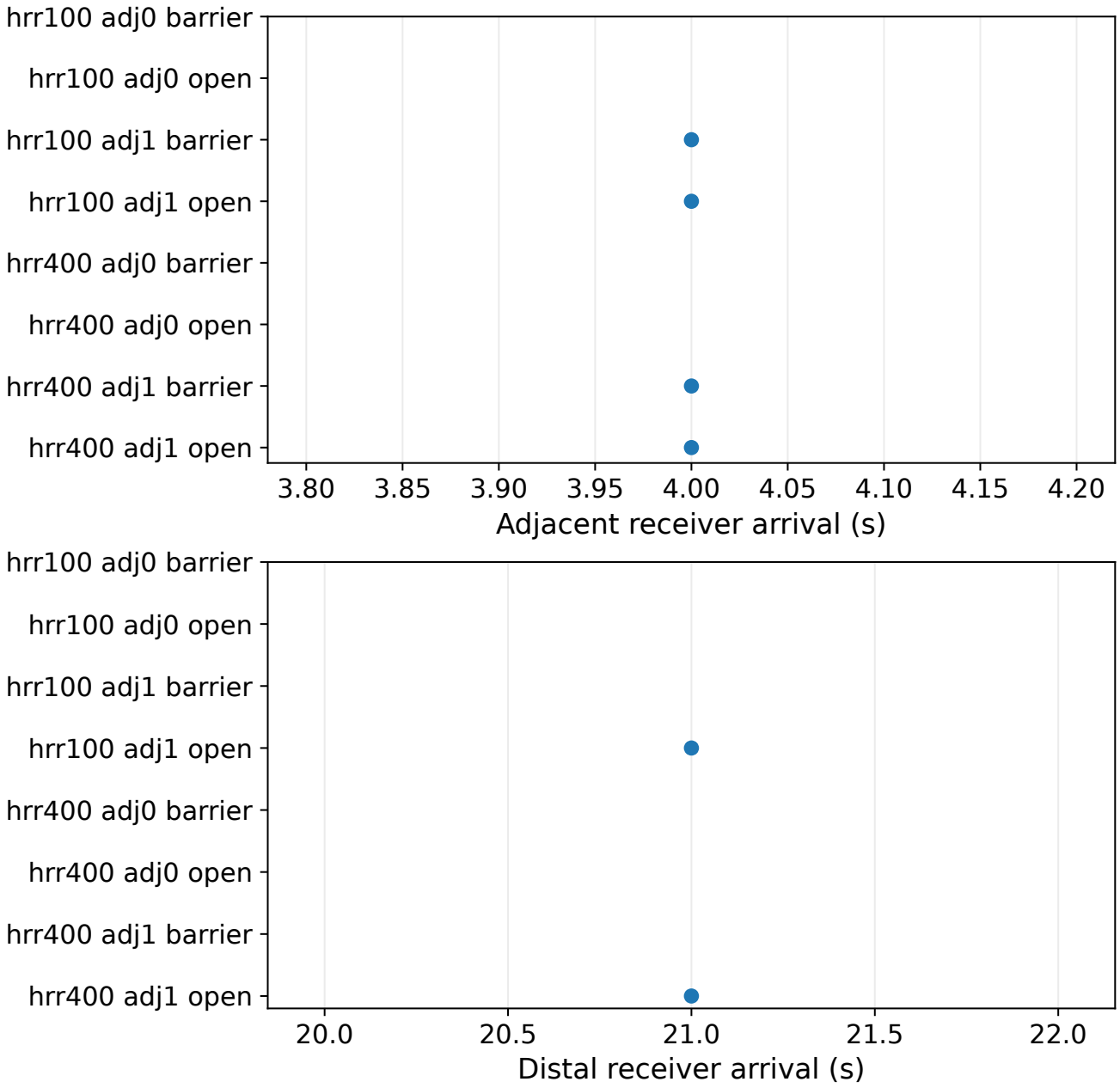


Figure 4: Adjacent and distal receiver arrival times for all eight simulation conditions. A missing observation is labeled no arrival, not plotted at zero. The selected simulation quantities are identified with the calculated metrics.

7 Calculated metrics and verification decision

Source HRR fourfold response error — PASS

Criterion	≤ 0.01
Observed	*Units: fraction.* 0

Source HRR control-repeat error — PASS

Criterion	≤ 0.001
Observed	*Units: fraction.* 0

Isolated receiver heat-exposure minimum — PASS

Criterion	> 0
Observed	*Units: kJ.* 438590

Isolated heat fourfold response error — PASS

Criterion	≤ 0.02
Observed	*Units: fraction.* 0

Isolated receiver arrivals — PASS

Criterion	0 of 8
Observed	*Units: count.* 0

Nonburnable barrier arrivals — PASS

Criterion	0 of 4
Observed	*Units: count.* 0

ADJ0 ordinary-path arrivals — PASS

Criterion	0 of 8 adjacent/distal checks
Observed	*Units: count.* 0

ADJ1 open-path arrivals — PASS

Criterion	4 of 4 adjacent/distal checks
Observed	*Units: count.* 4

Minimum ADJ1-open wildland spread rate — PASS

Criterion	$>$ 1e-4 in both open simulation conditions
Observed	*Units: m/min.* 70.237

Open-path TOA HRR-control difference — PASS	
Criterion	≤ 1
Observed	*Units: s.* 0

Intended heat-only urban-to-wildland ignition capability — NOT EVALUABLE	
Criterion	heat exposure can initiate a disconnected wildland receiver
Observed	*Units: capability.* no source-code state transition exists for FBFM != 91 target cells

Current-implementation characterization: **PASS**. Published intended heat-only U-to-W capability: **NOT EVALUABLE**. Overall case decision: **NOT EVALUABLE**. These two decision layers are deliberately not collapsed: agreement with the currently missing transition cannot verify that the documented capability has been implemented.

8 Technical interpretation

The implementation-characterization checks pass: source HRR and isolated received heat preserve the fourfold amplitude response, ordinary propagation responds to ADJ, and the nonburnable control remains intact. Yet the isolated wildland receiver has no arrival despite positive heat exposure. This combination demonstrates the distinction between heat deposition and a heat-driven ignition pathway.

The source selects heat-derived spread only for urban FBFM91 targets. Wildland receivers retain the ordinary surface-fire path. Agreement with this current behavior is a characterization PASS; it cannot establish the separately claimed heat-only urban-to-wildland capability. The overall capability verdict therefore remains NOT EVALUABLE. Missing TOA is shown as missing, never as an arrival at zero seconds.

9 Scope and limitations

This case covers deterministic DFC/radiation exposure from urban FBFM91 sources to FBFM1 receivers on a flat 10 m grid, two HRR amplitudes, an ADJ control, and one nonburnable barrier. It does not validate heat-transfer coefficients, material ignition thresholds, physical flame dimensions, firebrands, crown fire, topography, suppression, or stochastic behavior. It also does not claim that lack of a target law is desirable; it records that current-source fact separately from the published intended capability.

The supplied publications are technical evidence, not executable specifications. Their notation and default-value ambiguities, including combustible-fraction semantics and dimensional interpretation of the heat-derived ROS equations, require resolution before a future U-to-W ignition-time test can have a quantitative physical reference calculation. Spatial and temporal convergence are outside this case and belong to CASE48_WGC and CASE49_WTC, respectively.

References

[1] D. M. J. Purnomo et al., "Reconstructing modes of destruction in wildland–urban interface fires using a semi-physical level-set model," *Proceedings of the Combustion Institute*, 40 (2024), 105755. doi:10.1016/j.proci.2024.105755.

[2] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.